Test ID	Category	Command	UART Byte Sequence	Extra Input / Condition	Expected Trigger	Matrix 1	Matrix 2	Row / Shift	Column / Shift	Expected Data / Configuration	Expected UART Response	Description	Result
TC1.1	Reset	RESET	--	rst_n=0 for 5 rising edges; UART/data inputs initialized to 0	All operation-trigger outputs remain 0	0	0	0	0	matrix_signed=0; clamp_max=0; data_out=0; quantizer fields=0	uart_tx_new=0; uart_tx_data=0	Verify asynchronous reset initializes parser state and all registered outputs.	Pass
TC2.1	Randomized tests	QUANTIZE	7 random legal bytes	Random matrix 0-3, dtype 0-7, zero point 0-255, positive scale bit pattern	quantization_op=1 for 1 clock on byte 7	Random target	--	--	--	dtype, zero point, and 32-bit scale match transmitted fields	--	Original randomized QUANTIZE test.	Pass
TC2.2	Randomized tests	LOAD_ELEMENT	3 random legal bytes	Random matrix 0-3, row 0-3, column 0-3, data 0-255	data_load=1 for 1 clock on byte 3	Random target	--	Random row	Random column	data_out matches transmitted data	--	Original randomized LOAD_ELEMENT test.	Pass
TC2.3	Randomized tests	READ_ELEMENT	2 random legal bytes	Random matrix/address; data_in changed to a random byte after the request	data_read=1 for 1 clock on byte 2	Random source	--	Random row	Random column	--	uart_tx_new pulses 4 clocks after final byte; uart_tx_data=data_in	Original randomized READ_ELEMENT test.	Pass
TC2.4	Randomized tests	ADD	1 random legal byte	Random source matrices 0-3	arithmetic_op=1 for 1 clock	Random source 1	Random source 2	--	--	arithmetic_op_type=00	--	Original randomized ADD test.	Pass
TC2.5	Randomized tests	EL_MUL	1 random legal byte	Random source matrices 0-3	arithmetic_op=1 for 1 clock	Random source 1	Random source 2	--	--	arithmetic_op_type=01	--	Original randomized element-wise multiply test.	Pass
TC2.6	Randomized tests	MAT_MUL	1 random legal byte	Random source matrices 0-3	arithmetic_op=1 for 1 clock	Random source 1	Random source 2	--	--	arithmetic_op_type=11	--	Original randomized matrix multiply test.	Pass
TC2.7	Randomized tests	DOT_PROD	2 random legal bytes	Random source matrices and destination row/column 0-3	arithmetic_op=1 for 1 clock on byte 2	Random source 1	Random source 2	Random row	Random column	arithmetic_op_type=10	--	Original randomized dot-product test.	Pass
TC2.8	Randomized tests	CLEAR_MATRIX	1 random legal byte	Random matrix 0-3 and signed mode 0/1	matrix_config=1 for 1 clock	Random target	--	--	--	matrix_signed matches transmitted bit	--	Original randomized matrix-clear test.	Pass
TC2.9	Randomized tests	RELU	1 random legal byte	Random matrix 0-3	relu_op=1 for 1 clock	Random target	--	--	--	--	--	Original randomized ReLU test.	Pass
TC2.10	Randomized tests	CLAMP	2 random legal bytes	Random matrix 0-3 and clamp maximum 0-255	clamp_op=1 for 1 clock on byte 2	Random target	--	--	--	clamp_max matches transmitted byte	--	Original randomized clamp test.	Pass
TC2.11	Randomized tests	SHIFT_ROW_COL	2 random legal bytes	Random matrix 0-3, shift 0-3, and direction	Exactly one of shift_row/shift_col pulses on byte 2	Random target	--	Random shift	Random shift	Direction bit 1=row, 0=column	--	Original randomized row/column shift test.	Pass
TC3.1	QUANTIZE corner cases	QUANTIZE	10 00 00 00 00 00 00	Minimum field values	quantization_op=1 for 1 clock on final byte	0	--	--	--	dtype=0; zero_point=00; scale=00000000	--	Minimum target matrix, datatype, zero point, and scale.	Pass
TC3.2	QUANTIZE corner cases	QUANTIZE	1C E0 FF FF FF FF FF	Maximum raw field values	quantization_op=1 for 1 clock on final byte	3	--	--	--	dtype=7; zero_point=FF; scale=FFFFFFFF	--	Maximum target matrix, datatype, zero point, and raw scale bits.	Pass
TC3.3	QUANTIZE datatype coverage	QUANTIZE	10 00 80 3F 80 00 00	Datatype encoding 0; scale=1.0	quantization_op=1 for 1 clock on final byte	0	--	--	--	dtype=0; zero_point=80; scale=3F800000	--	Verify datatype value 0 is captured; destination matrix rotates as dtype mod 4.	Pass
TC3.4	QUANTIZE datatype coverage	QUANTIZE	14 20 80 3F 80 00 00	Datatype encoding 1; scale=1.0	quantization_op=1 for 1 clock on final byte	1	--	--	--	dtype=1; zero_point=80; scale=3F800000	--	Verify datatype value 1 is captured; destination matrix rotates as dtype mod 4.	Pass
TC3.5	QUANTIZE datatype coverage	QUANTIZE	18 40 80 3F 80 00 00	Datatype encoding 2; scale=1.0	quantization_op=1 for 1 clock on final byte	2	--	--	--	dtype=2; zero_point=80; scale=3F800000	--	Verify datatype value 2 is captured; destination matrix rotates as dtype mod 4.	Pass
TC3.6	QUANTIZE datatype coverage	QUANTIZE	1C 60 80 3F 80 00 00	Datatype encoding 3; scale=1.0	quantization_op=1 for 1 clock on final byte	3	--	--	--	dtype=3; zero_point=80; scale=3F800000	--	Verify datatype value 3 is captured; destination matrix rotates as dtype mod 4.	Pass
TC3.7	QUANTIZE datatype coverage	QUANTIZE	10 80 80 3F 80 00 00	Datatype encoding 4; scale=1.0	quantization_op=1 for 1 clock on final byte	0	--	--	--	dtype=4; zero_point=80; scale=3F800000	--	Verify datatype value 4 is captured; destination matrix rotates as dtype mod 4.	Pass

TC3.8	QUANTIZE datatype coverage	QUANTIZE	14 A0 80 3F 80 00 00	Datatype encoding 5; scale=1.0	quantization_op=1 for 1 clock on final byte	1	--	--	--	dtype=5; zero_point=80; scale=3F800000	--	Verify datatype value 5 is captured; destination matrix rotates as dtype mod 4.	Pass
TC3.9	QUANTIZE datatype coverage	QUANTIZE	18 C0 80 3F 80 00 00	Datatype encoding 6; scale=1.0	quantization_op=1 for 1 clock on final byte	2	--	--	--	dtype=6; zero_point=80; scale=3F800000	--	Verify datatype value 6 is captured; destination matrix rotates as dtype mod 4.	Pass
TC3.10	QUANTIZE datatype coverage	QUANTIZE	1C E0 80 3F 80 00 00	Datatype encoding 7; scale=1.0	quantization_op=1 for 1 clock on final byte	3	--	--	--	dtype=7; zero_point=80; scale=3F800000	--	Verify datatype value 7 is captured; destination matrix rotates as dtype mod 4.	Pass
TC3.11	QUANTIZE corner cases	QUANTIZE	18 00 80 BF 80 00 00	Negative IEEE-754 scale bit pattern (-1.0)	quantization_op=1 for 1 clock on final byte	2	--	--	--	dtype=0; zero_point=80; scale=BF800000	--	Verify the parser preserves the scale sign bit and all raw scale bits.	Pass
TC3.12	QUANTIZE corner cases	QUANTIZE	1C C0 A6 12 34 56 78	Distinct scale bytes	quantization_op=1 for 1 clock on final byte	3	--	--	--	dtype=6; zero_point=A6; scale=12345678	--	Verify big-endian byte assembly of the 32-bit scale field.	Pass
TC3.13	Reserved-bit tests	QUANTIZE	17 9F 55 3F 00 00 00	Reserved fields set to all 1s	quantization_op=1 for 1 clock on final byte	1	--	--	--	dtype=4; zero_point=55; scale=3F000000	--	Unused command bits must not change meaningful decoded fields.	Pass
TC4.1	LOAD_ELEMENT	LOAD_ELEMENT	20 00 00	Legal address/data	data_load=1 for 1 clock on byte 3	0	--	0	0	data_out=00	--	Minimum address and minimum data.	Pass
TC4.2	LOAD_ELEMENT	LOAD_ELEMENT	2C 33 FF	Legal address/data	data_load=1 for 1 clock on byte 3	3	--	3	3	data_out=FF	--	Maximum valid address and maximum data.	Pass
TC4.3	LOAD_ELEMENT	LOAD_ELEMENT	24 22 AA	Legal address/data	data_load=1 for 1 clock on byte 3	1	--	2	2	data_out=AA	--	First write to repeated address.	Pass
TC4.4	LOAD_ELEMENT	LOAD_ELEMENT	24 22 55	Legal address/data	data_load=1 for 1 clock on byte 3	1	--	2	2	data_out=55	--	Second write to the same address with replacement data.	Pass
TC4.5	LOAD_ELEMENT	LOAD_ELEMENT	2B 13 A5	Reserved bits in first byte set to 11	data_load=1 for 1 clock on byte 3	2	--	1	3	data_out=A5	--	Reserved bits set to 11.	Pass
TC5.1	READ_ELEMENT	READ_ELEMENT	30 00	data_in=00 before transmit response	data_read=1 for 1 clock on byte 2	0	--	0	0	--	uart_tx_new pulses 4 clocks after final byte; uart_tx_data=00	Minimum read address.	Pass
TC5.2	READ_ELEMENT	READ_ELEMENT	3C 33	data_in=FF before transmit response	data_read=1 for 1 clock on byte 2	3	--	3	3	--	uart_tx_new pulses 4 clocks after final byte; uart_tx_data=FF	Maximum valid read address.	Pass
TC5.3	READ_ELEMENT	READ_ELEMENT	3B 12	data_in=5A before transmit response	data_read=1 for 1 clock on byte 2	2	--	1	2	--	uart_tx_new pulses 4 clocks after final byte; uart_tx_data=5A	Reserved bits set to 11.	Pass
TC6.1	ADD exhaustive matrix pairs	ADD	40	All source-matrix combinations	arithmetic_op=1 for 1 clock	0	0	--	--	arithmetic_op_type=00	--	Verify ADD source pair (0,0).	Pass
TC6.2	ADD exhaustive matrix pairs	ADD	41	All source-matrix combinations	arithmetic_op=1 for 1 clock	0	1	--	--	arithmetic_op_type=00	--	Verify ADD source pair (0,1).	Pass
TC6.3	ADD exhaustive matrix pairs	ADD	42	All source-matrix combinations	arithmetic_op=1 for 1 clock	0	2	--	--	arithmetic_op_type=00	--	Verify ADD source pair (0,2).	Pass
TC6.4	ADD exhaustive matrix pairs	ADD	43	All source-matrix combinations	arithmetic_op=1 for 1 clock	0	3	--	--	arithmetic_op_type=00	--	Verify ADD source pair (0,3).	Pass
TC6.5	ADD exhaustive matrix pairs	ADD	44	All source-matrix combinations	arithmetic_op=1 for 1 clock	1	0	--	--	arithmetic_op_type=00	--	Verify ADD source pair (1,0).	Pass
TC6.6	ADD exhaustive matrix pairs	ADD	45	All source-matrix combinations	arithmetic_op=1 for 1 clock	1	1	--	--	arithmetic_op_type=00	--	Verify ADD source pair (1,1).	Pass
TC6.7	ADD exhaustive matrix pairs	ADD	46	All source-matrix combinations	arithmetic_op=1 for 1 clock	1	2	--	--	arithmetic_op_type=00	--	Verify ADD source pair (1,2).	Pass
TC6.8	ADD exhaustive matrix pairs	ADD	47	All source-matrix combinations	arithmetic_op=1 for 1 clock	1	3	--	--	arithmetic_op_type=00	--	Verify ADD source pair (1,3).	Pass
TC6.9	ADD exhaustive matrix pairs	ADD	48	All source-matrix combinations	arithmetic_op=1 for 1 clock	2	0	--	--	arithmetic_op_type=00	--	Verify ADD source pair (2,0).	Pass
TC6.10	ADD exhaustive matrix pairs	ADD	49	All source-matrix combinations	arithmetic_op=1 for 1 clock	2	1	--	--	arithmetic_op_type=00	--	Verify ADD source pair (2,1).	Pass
TC6.11	ADD exhaustive matrix pairs	ADD	4A	All source-matrix combinations	arithmetic_op=1 for 1 clock	2	2	--	--	arithmetic_op_type=00	--	Verify ADD source pair (2,2).	Pass
TC6.12	ADD exhaustive matrix pairs	ADD	4B	All source-matrix combinations	arithmetic_op=1 for 1 clock	2	3	--	--	arithmetic_op_type=00	--	Verify ADD source pair (2,3).	Pass

TC6.13	ADD exhaustive matrix pairs	ADD	4C	All source-matrix combinations	arithmetic_op=1 for 1 clock	3	0	--	--	arithmetic_op_type=00	--	Verify ADD source pair (3,0).	Pass
TC6.14	ADD exhaustive matrix pairs	ADD	4D	All source-matrix combinations	arithmetic_op=1 for 1 clock	3	1	--	--	arithmetic_op_type=00	--	Verify ADD source pair (3,1).	Pass
TC6.15	ADD exhaustive matrix pairs	ADD	4E	All source-matrix combinations	arithmetic_op=1 for 1 clock	3	2	--	--	arithmetic_op_type=00	--	Verify ADD source pair (3,2).	Pass
TC6.16	ADD exhaustive matrix pairs	ADD	4F	All source-matrix combinations	arithmetic_op=1 for 1 clock	3	3	--	--	arithmetic_op_type=00	--	Verify ADD source pair (3,3).	Pass
TC7.1	EL_MUL exhaustive matrix pairs	EL_MUL	50	All source-matrix combinations	arithmetic_op=1 for 1 clock	0	0	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (0,0).	Pass
TC7.2	EL_MUL exhaustive matrix pairs	EL_MUL	51	All source-matrix combinations	arithmetic_op=1 for 1 clock	0	1	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (0,1).	Pass
TC7.3	EL_MUL exhaustive matrix pairs	EL_MUL	52	All source-matrix combinations	arithmetic_op=1 for 1 clock	0	2	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (0,2).	Pass
TC7.4	EL_MUL exhaustive matrix pairs	EL_MUL	53	All source-matrix combinations	arithmetic_op=1 for 1 clock	0	3	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (0,3).	Pass
TC7.5	EL_MUL exhaustive matrix pairs	EL_MUL	54	All source-matrix combinations	arithmetic_op=1 for 1 clock	1	0	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (1,0).	Pass
TC7.6	EL_MUL exhaustive matrix pairs	EL_MUL	55	All source-matrix combinations	arithmetic_op=1 for 1 clock	1	1	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (1,1).	Pass
TC7.7	EL_MUL exhaustive matrix pairs	EL_MUL	56	All source-matrix combinations	arithmetic_op=1 for 1 clock	1	2	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (1,2).	Pass
TC7.8	EL_MUL exhaustive matrix pairs	EL_MUL	57	All source-matrix combinations	arithmetic_op=1 for 1 clock	1	3	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (1,3).	Pass
TC7.9	EL_MUL exhaustive matrix pairs	EL_MUL	58	All source-matrix combinations	arithmetic_op=1 for 1 clock	2	0	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (2,0).	Pass
TC7.10	EL_MUL exhaustive matrix pairs	EL_MUL	59	All source-matrix combinations	arithmetic_op=1 for 1 clock	2	1	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (2,1).	Pass
TC7.11	EL_MUL exhaustive matrix pairs	EL_MUL	5A	All source-matrix combinations	arithmetic_op=1 for 1 clock	2	2	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (2,2).	Pass
TC7.12	EL_MUL exhaustive matrix pairs	EL_MUL	5B	All source-matrix combinations	arithmetic_op=1 for 1 clock	2	3	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (2,3).	Pass
TC7.13	EL_MUL exhaustive matrix pairs	EL_MUL	5C	All source-matrix combinations	arithmetic_op=1 for 1 clock	3	0	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (3,0).	Pass
TC7.14	EL_MUL exhaustive matrix pairs	EL_MUL	5D	All source-matrix combinations	arithmetic_op=1 for 1 clock	3	1	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (3,1).	Pass
TC7.15	EL_MUL exhaustive matrix pairs	EL_MUL	5E	All source-matrix combinations	arithmetic_op=1 for 1 clock	3	2	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (3,2).	Pass
TC7.16	EL_MUL exhaustive matrix pairs	EL_MUL	5F	All source-matrix combinations	arithmetic_op=1 for 1 clock	3	3	--	--	arithmetic_op_type=01	--	Verify EL_MUL source pair (3,3).	Pass
TC8.1	MAT_MUL exhaustive matrix pairs	MAT_MUL	70	All source-matrix combinations	arithmetic_op=1 for 1 clock	0	0	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (0,0).	Pass
TC8.2	MAT_MUL exhaustive matrix pairs	MAT_MUL	71	All source-matrix combinations	arithmetic_op=1 for 1 clock	0	1	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (0,1).	Pass
TC8.3	MAT_MUL exhaustive matrix pairs	MAT_MUL	72	All source-matrix combinations	arithmetic_op=1 for 1 clock	0	2	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (0,2).	Pass
TC8.4	MAT_MUL exhaustive matrix pairs	MAT_MUL	73	All source-matrix combinations	arithmetic_op=1 for 1 clock	0	3	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (0,3).	Pass
TC8.5	MAT_MUL exhaustive matrix pairs	MAT_MUL	74	All source-matrix combinations	arithmetic_op=1 for 1 clock	1	0	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (1,0).	Pass
TC8.6	MAT_MUL exhaustive matrix pairs	MAT_MUL	75	All source-matrix combinations	arithmetic_op=1 for 1 clock	1	1	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (1,1).	Pass
TC8.7	MAT_MUL exhaustive matrix pairs	MAT_MUL	76	All source-matrix combinations	arithmetic_op=1 for 1 clock	1	2	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (1,2).	Pass
TC8.8	MAT_MUL exhaustive matrix pairs	MAT_MUL	77	All source-matrix combinations	arithmetic_op=1 for 1 clock	1	3	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (1,3).	Pass
TC8.9	MAT_MUL exhaustive matrix pairs	MAT_MUL	78	All source-matrix combinations	arithmetic_op=1 for 1 clock	2	0	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (2,0).	Pass
TC8.10	MAT_MUL exhaustive matrix pairs	MAT_MUL	79	All source-matrix combinations	arithmetic_op=1 for 1 clock	2	1	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (2,1).	Pass
TC8.11	MAT_MUL exhaustive matrix pairs	MAT_MUL	7A	All source-matrix combinations	arithmetic_op=1 for 1 clock	2	2	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (2,2).	Pass

TC8.12	MAT_MUL exhaustive matrix pairs	MAT_MUL	7B	All source-matrix combinations	arithmetic_op=1 for 1 clock	2	3	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (2,3).	Pass
TC8.13	MAT_MUL exhaustive matrix pairs	MAT_MUL	7C	All source-matrix combinations	arithmetic_op=1 for 1 clock	3	0	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (3,0).	Pass
TC8.14	MAT_MUL exhaustive matrix pairs	MAT_MUL	7D	All source-matrix combinations	arithmetic_op=1 for 1 clock	3	1	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (3,1).	Pass
TC8.15	MAT_MUL exhaustive matrix pairs	MAT_MUL	7E	All source-matrix combinations	arithmetic_op=1 for 1 clock	3	2	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (3,2).	Pass
TC8.16	MAT_MUL exhaustive matrix pairs	MAT_MUL	7F	All source-matrix combinations	arithmetic_op=1 for 1 clock	3	3	--	--	arithmetic_op_type=11	--	Verify MAT_MUL source pair (3,3).	Pass
TC9.1	DOT_PROD source pairs	DOT_PROD	60 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	0	0	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (0,0) at destination [0,0].	Pass
TC9.2	DOT_PROD source pairs	DOT_PROD	61 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	0	1	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (0,1) at destination [0,0].	Pass
TC9.3	DOT_PROD source pairs	DOT_PROD	62 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	0	2	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (0,2) at destination [0,0].	Pass
TC9.4	DOT_PROD source pairs	DOT_PROD	63 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	0	3	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (0,3) at destination [0,0].	Pass
TC9.5	DOT_PROD source pairs	DOT_PROD	64 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	1	0	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (1,0) at destination [0,0].	Pass
TC9.6	DOT_PROD source pairs	DOT_PROD	65 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	1	1	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (1,1) at destination [0,0].	Pass
TC9.7	DOT_PROD source pairs	DOT_PROD	66 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	1	2	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (1,2) at destination [0,0].	Pass
TC9.8	DOT_PROD source pairs	DOT_PROD	67 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	1	3	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (1,3) at destination [0,0].	Pass
TC9.9	DOT_PROD source pairs	DOT_PROD	68 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	2	0	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (2,0) at destination [0,0].	Pass
TC9.10	DOT_PROD source pairs	DOT_PROD	69 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	2	1	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (2,1) at destination [0,0].	Pass
TC9.11	DOT_PROD source pairs	DOT_PROD	6A 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	2	2	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (2,2) at destination [0,0].	Pass
TC9.12	DOT_PROD source pairs	DOT_PROD	6B 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	2	3	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (2,3) at destination [0,0].	Pass
TC9.13	DOT_PROD source pairs	DOT_PROD	6C 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	3	0	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (3,0) at destination [0,0].	Pass
TC9.14	DOT_PROD source pairs	DOT_PROD	6D 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	3	1	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (3,1) at destination [0,0].	Pass
TC9.15	DOT_PROD source pairs	DOT_PROD	6E 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	3	2	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (3,2) at destination [0,0].	Pass
TC9.16	DOT_PROD source pairs	DOT_PROD	6F 00	Destination row=0, column=0	arithmetic_op=1 for 1 clock on byte 2	3	3	0	0	arithmetic_op_type=10	--	Verify dot-product source pair (3,3) at destination [0,0].	Pass
TC9.17	DOT_PROD corner case	DOT_PROD	6C 33	Maximum destination row and column	arithmetic_op=1 for 1 clock on byte 2	3	0	3	3	arithmetic_op_type=10	--	Verify destination address maximum and source pair (3,0).	Pass
TC10.1	CLEAR_MATRIX	CLEAR_MATRIX	80	signed_mode=0	matrix_config=1 for 1 clock	0	--	--	--	matrix_signed=0	--	Clear matrix 0 in unsigned mode.	Pass
TC10.2	CLEAR_MATRIX	CLEAR_MATRIX	82	signed_mode=1	matrix_config=1 for 1 clock	0	--	--	--	matrix_signed=1	--	Clear matrix 0 in signed mode.	Pass
TC10.3	CLEAR_MATRIX	CLEAR_MATRIX	84	signed_mode=0	matrix_config=1 for 1 clock	1	--	--	--	matrix_signed=0	--	Clear matrix 1 in unsigned mode.	Pass
TC10.4	CLEAR_MATRIX	CLEAR_MATRIX	86	signed_mode=1	matrix_config=1 for 1 clock	1	--	--	--	matrix_signed=1	--	Clear matrix 1 in signed mode.	Pass
TC10.5	CLEAR_MATRIX	CLEAR_MATRIX	88	signed_mode=0	matrix_config=1 for 1 clock	2	--	--	--	matrix_signed=0	--	Clear matrix 2 in unsigned mode.	Pass
TC10.6	CLEAR_MATRIX	CLEAR_MATRIX	8A	signed_mode=1	matrix_config=1 for 1 clock	2	--	--	--	matrix_signed=1	--	Clear matrix 2 in signed mode.	Pass
TC10.7	CLEAR_MATRIX	CLEAR_MATRIX	8C	signed_mode=0	matrix_config=1 for 1 clock	3	--	--	--	matrix_signed=0	--	Clear matrix 3 in unsigned mode.	Pass
TC10.8	CLEAR_MATRIX	CLEAR_MATRIX	8E	signed_mode=1	matrix_config=1 for 1 clock	3	--	--	--	matrix_signed=1	--	Clear matrix 3 in signed mode.	Pass
TC11.1	RELU	RELU	90	Reserved bits=00	relu_op=1 for 1 clock	0	--	--	--	--	--	Apply ReLU to matrix 0.	Pass

TC11.2	RELU	RELU	94	Reserved bits=00	relu_op=1 for 1 clock	1	--	--	--	--	--	Apply ReLU to matrix 1.	Pass
TC11.3	RELU	RELU	98	Reserved bits=00	relu_op=1 for 1 clock	2	--	--	--	--	--	Apply ReLU to matrix 2.	Pass
TC11.4	RELU	RELU	9C	Reserved bits=00	relu_op=1 for 1 clock	3	--	--	--	--	--	Apply ReLU to matrix 3.	Pass
TC11.5	Reserved-bit tests	RELU	9B	Reserved bits=11	relu_op=1 for 1 clock	2	--	--	--	--	--	Reserved low bits must not affect ReLU matrix selection.	Pass
TC12.1	CLAMP boundaries	CLAMP	A0 00	clamp_max=00	clamp_op=1 for 1 clock on byte 2	0	--	--	--	clamp_max=00	--	Apply clamp boundary value 00 to matrix 0.	Pass
TC12.2	CLAMP boundaries	CLAMP	A0 FF	clamp_max=FF	clamp_op=1 for 1 clock on byte 2	0	--	--	--	clamp_max=FF	--	Apply clamp boundary value FF to matrix 0.	Pass
TC12.3	CLAMP boundaries	CLAMP	A4 00	clamp_max=00	clamp_op=1 for 1 clock on byte 2	1	--	--	--	clamp_max=00	--	Apply clamp boundary value 00 to matrix 1.	Pass
TC12.4	CLAMP boundaries	CLAMP	A4 FF	clamp_max=FF	clamp_op=1 for 1 clock on byte 2	1	--	--	--	clamp_max=FF	--	Apply clamp boundary value FF to matrix 1.	Pass
TC12.5	CLAMP boundaries	CLAMP	A8 00	clamp_max=00	clamp_op=1 for 1 clock on byte 2	2	--	--	--	clamp_max=00	--	Apply clamp boundary value 00 to matrix 2.	Pass
TC12.6	CLAMP boundaries	CLAMP	A8 FF	clamp_max=FF	clamp_op=1 for 1 clock on byte 2	2	--	--	--	clamp_max=FF	--	Apply clamp boundary value FF to matrix 2.	Pass
TC12.7	CLAMP boundaries	CLAMP	AC 00	clamp_max=00	clamp_op=1 for 1 clock on byte 2	3	--	--	--	clamp_max=00	--	Apply clamp boundary value 00 to matrix 3.	Pass
TC12.8	CLAMP boundaries	CLAMP	AC FF	clamp_max=FF	clamp_op=1 for 1 clock on byte 2	3	--	--	--	clamp_max=FF	--	Apply clamp boundary value FF to matrix 3.	Pass
TC12.9	Reserved-bit tests	CLAMP	A7 7F	Reserved bits=11	clamp_op=1 for 1 clock on byte 2	1	--	--	--	clamp_max=7F	--	Reserved bits must not affect matrix selection or clamp value.	Pass
TC13.1	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	B0 00	direction=column	shift_col=1 for 1 clock; shift_row=0	0	--	0	0	Decoded shift amount=0	--	Column shift by minimum amount 0. Matrix 0.	Pass
TC13.2	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	B0 30	direction=column	shift_col=1 for 1 clock; shift_row=0	0	--	3	3	Decoded shift amount=3	--	Column shift by maximum valid amount 3. Matrix 0.	Pass
TC13.3	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	B0 08	direction=row	shift_row=1 for 1 clock; shift_col=0	0	--	0	0	Decoded shift amount=0	--	Row shift by minimum amount 0. Matrix 0.	Pass
TC13.4	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	B0 38	direction=row	shift_row=1 for 1 clock; shift_col=0	0	--	3	3	Decoded shift amount=3	--	Row shift by maximum valid amount 3. Matrix 0.	Pass
TC13.5	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	B4 00	direction=column	shift_col=1 for 1 clock; shift_row=0	1	--	0	0	Decoded shift amount=0	--	Column shift by minimum amount 0. Matrix 1.	Pass
TC13.6	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	B4 30	direction=column	shift_col=1 for 1 clock; shift_row=0	1	--	3	3	Decoded shift amount=3	--	Column shift by maximum valid amount 3. Matrix 1.	Pass
TC13.7	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	B4 08	direction=row	shift_row=1 for 1 clock; shift_col=0	1	--	0	0	Decoded shift amount=0	--	Row shift by minimum amount 0. Matrix 1.	Pass
TC13.8	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	B4 38	direction=row	shift_row=1 for 1 clock; shift_col=0	1	--	3	3	Decoded shift amount=3	--	Row shift by maximum valid amount 3. Matrix 1.	Pass
TC13.9	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	B8 00	direction=column	shift_col=1 for 1 clock; shift_row=0	2	--	0	0	Decoded shift amount=0	--	Column shift by minimum amount 0. Matrix 2.	Pass
TC13.10	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	B8 30	direction=column	shift_col=1 for 1 clock; shift_row=0	2	--	3	3	Decoded shift amount=3	--	Column shift by maximum valid amount 3. Matrix 2.	Pass
TC13.11	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	B8 08	direction=row	shift_row=1 for 1 clock; shift_col=0	2	--	0	0	Decoded shift amount=0	--	Row shift by minimum amount 0. Matrix 2.	Pass
TC13.12	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	B8 38	direction=row	shift_row=1 for 1 clock; shift_col=0	2	--	3	3	Decoded shift amount=3	--	Row shift by maximum valid amount 3. Matrix 2.	Pass
TC13.13	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	BC 00	direction=column	shift_col=1 for 1 clock; shift_row=0	3	--	0	0	Decoded shift amount=0	--	Column shift by minimum amount 0. Matrix 3.	Pass
TC13.14	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	BC 30	direction=column	shift_col=1 for 1 clock; shift_row=0	3	--	3	3	Decoded shift amount=3	--	Column shift by maximum valid amount 3. Matrix 3.	Pass
TC13.15	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	BC 08	direction=row	shift_row=1 for 1 clock; shift_col=0	3	--	0	0	Decoded shift amount=0	--	Row shift by minimum amount 0. Matrix 3.	Pass
TC13.16	SHIFT_ROW_COL boundaries	SHIFT_ROW_COL	BC 38	direction=row	shift_row=1 for 1 clock; shift_col=0	3	--	3	3	Decoded shift amount=3	--	Row shift by maximum valid amount 3. Matrix 3.	Pass
TC13.17	Reserved-bit tests	SHIFT_ROW_COL	BB 2F	First-byte reserved bits=11; second-byte reserved bits=111	shift_row=1 for 1 clock; shift_col=0	2	--	2	2	Decoded shift amount=2	--	Reserved bits must not affect row-shift decoding.	Pass

TC14.1	Invalid opcode handling	INVALID 0x0	00	uart_rx_new asserted for one byte	No supported operation-trigger output may pulse	Unchanged	Unchanged	Unchanged	Unchanged	No valid command decode	No UART response	Verify unsupported command byte 00 is ignored.	Pass
TC14.2	Invalid opcode handling	INVALID 0xC	C0	uart_rx_new asserted for one byte	No supported operation-trigger output may pulse	Unchanged	Unchanged	Unchanged	Unchanged	No valid command decode	No UART response	Verify unsupported command byte C0 is ignored.	Pass
TC14.3	Invalid opcode handling	INVALID 0xD	D5	uart_rx_new asserted for one byte	No supported operation-trigger output may pulse	Unchanged	Unchanged	Unchanged	Unchanged	No valid command decode	No UART response	Verify unsupported command byte D5 is ignored.	Pass
TC14.4	Invalid opcode handling	INVALID 0xE	EA	uart_rx_new asserted for one byte	No supported operation-trigger output may pulse	Unchanged	Unchanged	Unchanged	Unchanged	No valid command decode	No UART response	Verify unsupported command byte EA is ignored.	Pass
TC14.5	Invalid opcode handling	INVALID 0xF	FF	uart_rx_new asserted for one byte	No supported operation-trigger output may pulse	Unchanged	Unchanged	Unchanged	Unchanged	No valid command decode	No UART response	Verify unsupported command byte FF is ignored.	Pass
TC15.1	Reset recovery	PARTIAL QUANTIZE + RESET + ADD	14 40 20; reset; 43	Assert rst_n=0 for 3 clocks after three QUANTIZE bytes, then release for 3 clocks	Partial QUANTIZE produces no quantization_op; post-reset ADD produces arithmetic_op=1	0 after ADD	3 after ADD	0 after reset	0 after reset	All registered fields reset; post-reset arithmetic_op_type=00	uart_tx_new=0	Verify an incomplete multi-byte command is discarded and the parser correctly accepts a new command after reset.	Pass
TC16.1	Ignore data when uart_rx_new=0	NO COMMAND	FF then 00	uart_rx_new=0 throughout; each value held for 3 clocks	No operation-trigger output may pulse; parser must not execute a command.	No command update expected	No command update expected	--	--	Parser state remains idle	No UART response	Verify data-bus activity is ignored when uart_rx_new is low.	Pass